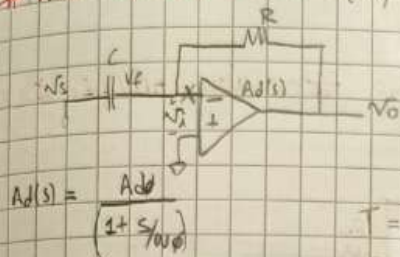


$$R_x^0 = \frac{1 + (R_5 \parallel R_4) \left(\frac{1}{R_2 \parallel R_{T2}} + g_{m2} \right)}{\frac{1}{R_2 \parallel R_{T2}}} = R_2 \parallel R_{T2} + (R_5 \parallel R_4) \underbrace{(1 + g_{m2} (R_2 \parallel R_{T2}))}_{\text{Miller effect}}$$

$$R_x^0 = 394.2 \text{ K}\Omega \quad \checkmark$$

$$f_A = 20.19 \text{ KHz} \quad \checkmark$$

TIME DIFFERENTIATOR: (fully compensated amplifier)



DIRECT LOOP INSPECTION: no loading effects because it is a CRMP

$$T = \frac{V_o}{V_i}$$

$$\begin{aligned} V_o &= -V_i A_d(s) \\ V_i &= \frac{1}{sC} V_o = \frac{z_1}{sC} V_o \end{aligned}$$

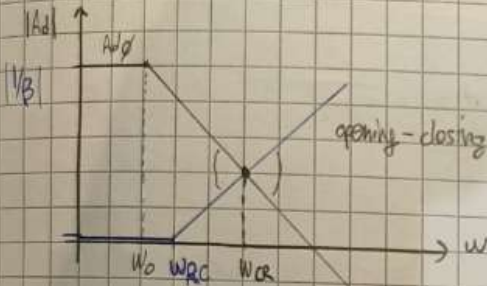
$$T = \frac{z_1}{z_1 + z_2} A_d(s) = \frac{1}{1 + sCR} A_d(s)$$

$$\begin{aligned} z_1 &= \frac{1}{sC} \\ z_2 &= R \end{aligned}$$

$$T = \frac{B(s)}{(1 + sCR)} \cdot \frac{A_d(s)}{(1 + \frac{s}{\omega_0})}$$

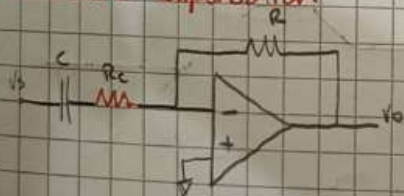
stability problems

$$\omega_{RC} = \frac{1}{CR}$$



opening-closing ratio of $\pm 40 \text{ dB} \rightarrow$ low PM $\rightarrow$ stability problems

we need a zero in the $p(s) \rightarrow$ let's compensate
series resistance compensation



$$T = \frac{z_1}{z_1 + z_2} A_d(s) = \frac{B_c(s)}{1 + sC(R + R_c)} A_d(s)$$

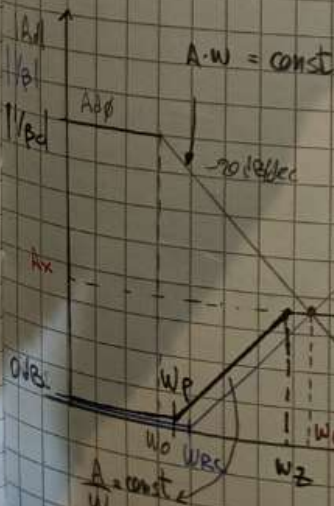
$\rightarrow$ we added a zero

$$\begin{aligned} z_1 &= \frac{1}{sC} + R_c \\ z_2 &= R \end{aligned}$$

$$T = \frac{1 + sCR_c}{1 + sC(R + R_c)} \cdot \frac{A_d(s)}{1 + \frac{s}{\omega_0}}$$

$$\omega_z = \frac{1}{CR_c}$$

$$\omega_p = \frac{1}{C(R + R_c)}$$



we move ω_{pc} a little to the left and since $A_f \approx 1$ when $T \gg 1$ we are changing the gain $\rightarrow$ not desired: to avoid this we should look for the conditions $\omega_p \approx \omega_{RC}$ and $\omega_z \approx \omega_{RC}$

$$\omega_p = \frac{1}{C(R + R_c)} \approx \frac{1}{CR} = \omega_{RC} \rightarrow R_c \ll R \quad \text{and} \quad \omega_z = \frac{1}{CR_c} \approx \omega_{RC} \rightarrow R_c = \frac{1}{C \omega_{RC}}$$

but what is the value of ω_{RC} ? We use the GAIN PRODUCT BANDWIDTH

$$\begin{aligned} A_{dp} \cdot \omega_0 &= A_x \cdot \omega_{RC} \quad \leftarrow -20 \text{ dB/decade} \\ \frac{1}{\omega_p} &= \frac{A_x}{\omega_{RC}} \quad \leftarrow +20 \text{ dB/decade} \end{aligned}$$

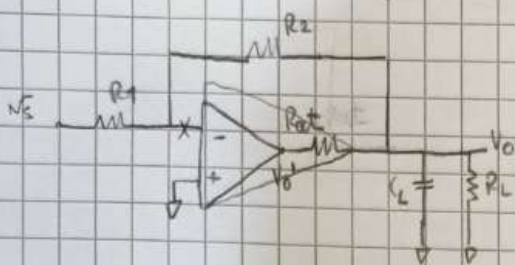
$$\omega_{RC} = \sqrt{A_{dp} \omega_0 \omega_p} \rightarrow R_c = \frac{1}{C \omega_{RC}}$$

① original design is not affected at all

capacitive load (fully compensated amplifier)

$$A_d(s) = A_{d0} / (1 + s/\omega_0)$$

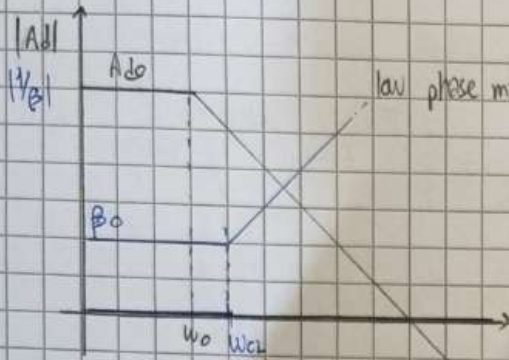
capacitive load implies that R_{out} of the opamp is no longer negligible



$$\beta(s) = \frac{v_i}{v_o} = \frac{v_i}{v_o} \cdot \frac{v_o}{v_o'} = \frac{R_1}{R_1 + R_2} \cdot \frac{R_L \parallel (R_1 + R_2)}{R_{out} + R_L \parallel (R_1 + R_2)} \cdot \frac{1}{1 + s C_L Z_L \parallel (R_1 + R_2)}$$

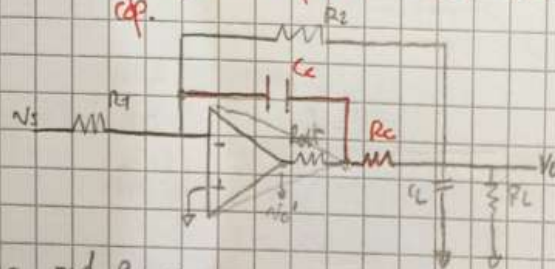
$$R_{CL} = R_L \parallel R_{out} \parallel (R_1 + R_2)$$

$$\omega_{CL} = \frac{1}{C_L R_{CL}} \quad \beta_0 = \frac{R_1}{R_1 + R_2} \cdot \frac{R_L'}{R_{out} + R_L'} \quad \beta(s) = \frac{\beta_0}{1 + s/\omega_{CL}}$$



low phase margin $\rightarrow$ delay $\rightarrow$ we need a faster response: compensation

making the response faster with a new path with small cap.



very complex pole network so let's study β_{CL} and $\beta_{CL\infty}$ to understand the behaviour

$$\beta_{CL} = \frac{R_1}{R_1 + R_2} \cdot \frac{R_L'}{R_{out} + R_C + R_L'} \approx \beta_0 \text{ to maintain the same behaviour of } \beta \text{ so } R_C \ll R_{out}$$

$$\beta_{CL\infty} = \frac{v_i}{v_o'} = \frac{R_1 \parallel R_2 \parallel R_C}{R_{out} + R_1 \parallel R_2 \parallel R_C} \approx \frac{R_C}{R_{out} + R_C} \approx \frac{R_C}{R_{out}}$$

$\downarrow$
CL shorted
 C_C shorted

ω_{PL} is the low frequency pole of the β_{CL} network and ω_{PH} is the high frequency pole of the β_{CL} network

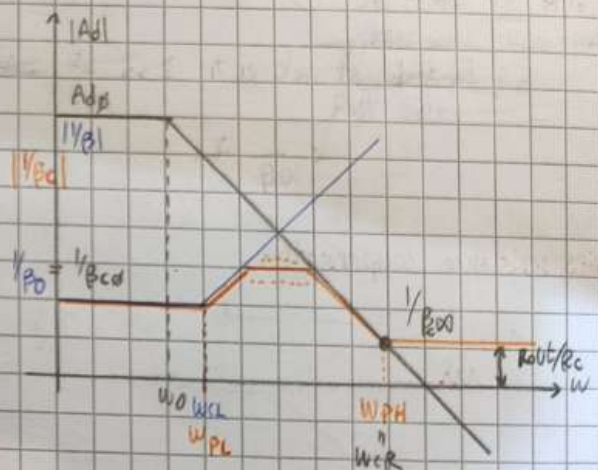
$$A_{d0} \cdot \omega_0 = \frac{1}{\beta_{CL\infty}} \cdot \omega_{PH} \quad \text{and } \omega_{CL} = \omega_{PH}$$

$$\omega_{PL} \approx \omega_{CL}$$

$$\omega_{PL} = \frac{1}{C_L R_1 \parallel (R_{out} + R_C) \parallel (R_1 + R_2)} \approx \omega_{CL}$$

$$\omega_{PH} = \frac{1}{C_C R_{CC}} = \frac{1}{C_C \cdot [R_1 \parallel R_2 + (R_C \parallel R_{out})]} \rightarrow C_C = \frac{1}{\omega_{CL} \cdot R_{CC}}$$

$\downarrow$
CL shorted
(LF)



- PROCEDURE: 1) we set ω_{CL} where we want some good phase margin (where we want to improve PM)
- 2) we derive $\beta_{CL\infty}$ and so R_C and verify $R_C \ll R_{out}$ so to have $\omega_{PL} \approx \omega_{CL}$ and $\beta_{CL} \approx \beta_0$
 - 3) we found C_C
 - 4) verify β_{CL} network works properly

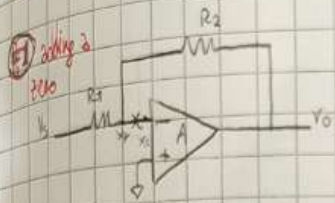
low frequency response is mainly regulated by a CM capacitor

by feedback because we can't consider those two parts separately and then we sum up

trade off bandwidth - PM

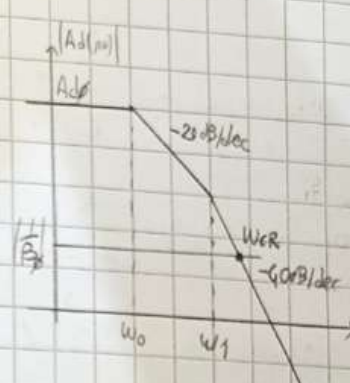
$\Delta\phi_m > 45^\circ \rightarrow W_z = W_{CR}$
 $\Delta\phi_m > 60^\circ \rightarrow W_z = \frac{W_{CR}}{4}$
 $W_z = 4 W_{CR}$

PARTIALLY COMPENSATED OPAmp STRATEGY :



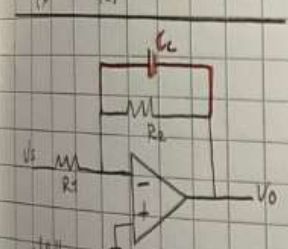
$$A_d(s) = \frac{A_{d0}}{(1 + s/\omega_0)(1 + s/\omega_1)}$$

if the gain is below a certain value we have a very low phase margin

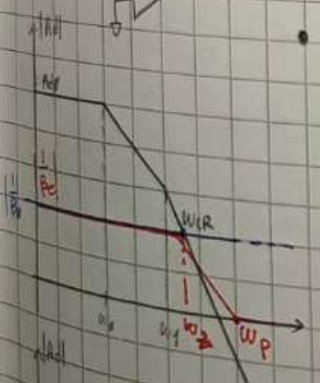


$$\frac{V_o}{V_i} = \frac{V_o}{-V_o} \cdot \frac{R_1}{R_1 + R_2} = A_d(s) \cdot \frac{R_1}{R_1 + R_2}$$

if $\frac{1}{\beta} = \frac{R_1 + R_2}{R_1} < \text{threshold} \rightarrow \text{very low phase margin} \rightarrow \text{compensation: ADDING A ZERO APPROACH}$ the W_{CR} with -20 dB/dec reducing the opening and closing voltages



$$\beta_c = \frac{R_1}{R_1 + R_2 \parallel \frac{1}{sC_c}} = \frac{R_1(1 + sR_2C_c)}{R_1(sR_2C_c + 1) + R_2} = \frac{R_1}{R_1 + R_2} \cdot \frac{(1 + sR_2C_c)}{(1 + s \frac{R_2R_1C_c}{R_2 + R_1})} = \frac{R_1}{R_1 + R_2} \cdot \frac{(1 + s/\omega_z)}{(1 + s/\omega_p)}$$



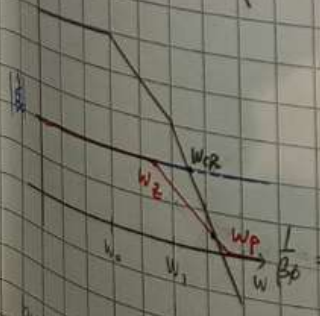
if we want $\Delta\phi = +45^\circ$ we can place $W_z = W_{CR}$: in this way we don't move the W_{CR} and the gain is almost equal.

$\Delta\phi = +45^\circ \rightarrow \text{can be used if } \frac{1}{\beta} \geq 10$

$W_z = W_{CR}$
 $A_d W_z = W_1 A_1$
 $A_1 W_1^2 = \frac{1}{\beta_0} W_{CR}^2 \rightarrow W_{CR} = \sqrt{\beta_0 W_1 W_0 A_d}$

$$W_z = W_{CR} = \frac{1}{R_2 C_c} \rightarrow C_c = \frac{1}{R_2 W_{CR}}$$

$$W_p = \frac{R_1 + R_2}{R_1 R_2 C_c} \text{ (don't care where } W_p \text{ is but having at least } 1 \geq 10 \text{ ensures it is)}$$



if we want $\Delta\phi = +60^\circ$ we should place $W_z = \frac{W_{CR}}{4}$ and $W_p = 4 W_{CR}$

$\rightarrow \text{can be used if } \frac{1}{\beta} \geq 16$
 the new W_{CR} will be a little different and the gain will be reduced
 $W_z = \frac{W_{CR}}{4} = \frac{1}{R_2 C_c} \rightarrow C_c = \frac{4}{R_2 W_{CR}}$
 $\frac{1}{\beta} = \frac{R_1 + R_2}{R_1} = 1 + \frac{R_2}{R_1}$

since we can decide only C_c we can set only W_z or W_p (generally W_z) and the rest of at least 16 = 1 come from the fact that we must ensure enough space to allow W_p to go far away from W_z since at $W_p, \frac{1}{\beta} \rightarrow \frac{1}{\beta_0} = 1 = 0 \text{ dB}$

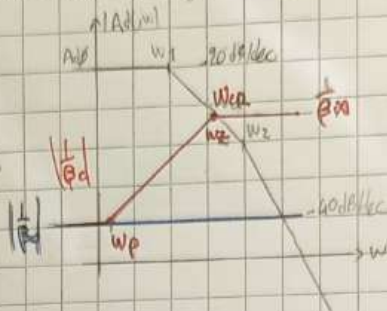
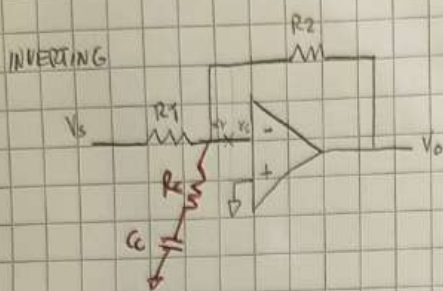
higher $\Delta\phi$ is paid in terms of gain so there is a minimum gain for which we can obtain a certain PM
 $\Delta\phi = +45^\circ$ if $\frac{1}{\beta} \geq 10$
 $\Delta\phi = +60^\circ$ if $\frac{1}{\beta} \geq 16$
 $\rightarrow$ For lower gains we must use NOISE GAIN COMPENSATION because W_z and W_p will get too close and there will be not enough space to ensure good $\Delta\phi$, additional phase becomes zero

#2

NOISE GAIN COMPENSATION

this can be used when we need high phase margin $\Delta\phi = 45^\circ$ or $\Delta\phi = 60^\circ$ and we have very low gain $\frac{1}{\beta} < 10$

can be used both with inverting and non-inverting configuration of opamps



$$T = \frac{-X_r}{X_s} = \frac{A_d(s)}{V_o} \cdot \frac{V_o}{R_2 + (R_1 + \frac{1}{sC})} = \dots$$

$$\beta_0 = \frac{R_1}{R_1 + R_2} \rightarrow \frac{1}{\beta_0} = 1 + \frac{R_2}{R_1}$$

if $\frac{R_2}{R_1} = 100 \rightarrow \frac{1}{\beta_0} \text{ small} \rightarrow \text{need noise gain compensation}$

we don't need to evaluate the complete β_c but just β_0 and β_{∞} . we want $\frac{1}{\beta_c}$ to cross at -20 dB/dec

$$\beta_{\infty} = \frac{R_1 // R_c}{R_1 // R_c + R_2} \quad \text{and} \quad \beta_c = \beta_0 \cdot \frac{1 + sR_c C_c}{1 + s/w_p} \rightarrow w_z = \frac{1}{R_c C_c}$$

we can set $w_z = w_{cr} \rightarrow \Delta\phi = 45^\circ$

or we can set $w_z = w_{cr}$ and $w_{cr} = \frac{w_z}{4} \rightarrow \Delta\phi = 60^\circ$

we decide w_{cr} depending on the phase margin we need! (1°) set w_{cr}

$$A_d w_1 = w_{cr} \frac{1}{\beta_0} \rightarrow \beta_{\infty} = \frac{R_1 // R_c}{R_1 // R_c + R_2} \rightarrow \frac{A_d w_1}{w_{cr}} = \frac{1 + R_2}{R_1 // R_c} \rightarrow R_1 // R_c = \frac{R_2}{\frac{A_d w_1}{w_{cr}} - 1} \rightarrow R_c = \frac{1}{\frac{1}{R_1 // R_c} - \frac{1}{R_1}}$$

$$w_z = \frac{1}{R_c C_c} \quad (3^\circ) \text{ find } w_z$$

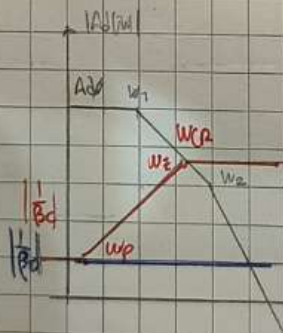
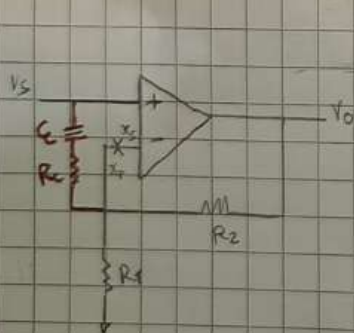
$$(4^\circ) w_p = \frac{1}{C_c R_c^{sc}} = \frac{1}{C_c (R_c + R_1 // R_2)}$$

if pole, zero method

$$(2^\circ) \text{ find } R_c$$

(5°) check if the PM is good

NON-INVERTING



$$T = \frac{-X_r}{X_s} = \frac{A_d(s)}{V_o} \cdot \frac{V_o}{R_1 // R_c + \frac{1}{sC}} = A_d(s) \cdot \beta_c$$

$$\beta_0 = \frac{R_1}{R_1 + R_2} \rightarrow \frac{1}{\beta_0} = 1 + \frac{R_2}{R_1}$$

$$\beta_{\infty} = \frac{R_1 // R_c}{R_1 // R_c + R_2} \rightarrow \frac{1}{\beta_{\infty}} = 1 + \frac{R_2}{R_1 // R_c}$$

$$\beta_c = \beta_0 \cdot \frac{1 + s/w_z}{1 + s/w_p} \rightarrow w_z = \frac{1}{R_c C_c}$$

(1°) set w_{cr} to have $\Delta\phi = 45^\circ$

$\Delta\phi = 60^\circ$

(2°) find R_c from $A_d w_1 = w_{cr} \frac{1}{\beta_0}$

$$(3^\circ) w_z = \frac{1}{R_c C_c}$$

$$(4^\circ) w_p = \frac{1}{C_c R_c^{sc}} = \frac{1}{C_c (R_c + R_1 // R_2)}$$

(5°) verify the result